On the Dynamic Regret of FTRL: Optimism with History Pruning

Naram Mhaisen and George Iosifidis

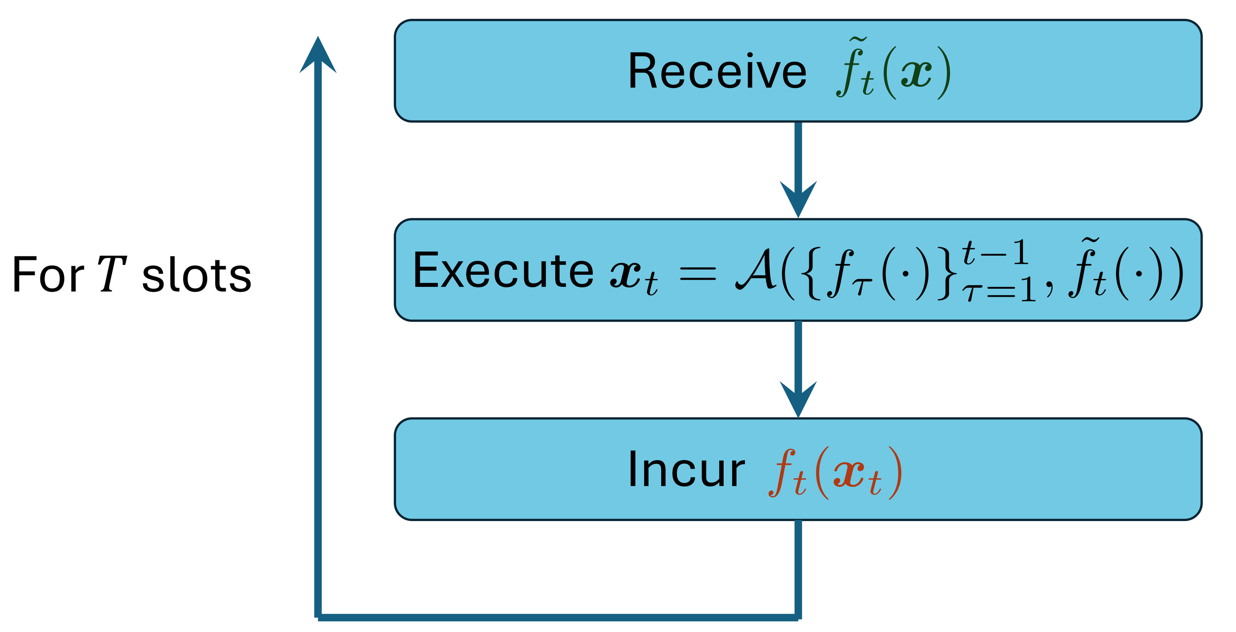
TU Delft — Delft University of Technology

Faculty of Electrical Engineering, Mathematics, and Computer Science.
Department of Software Technology, Networked Systems Group.

ICML International Conference On Machine Learning

1 Problem Setup

Online Convex Optimization with Hints :



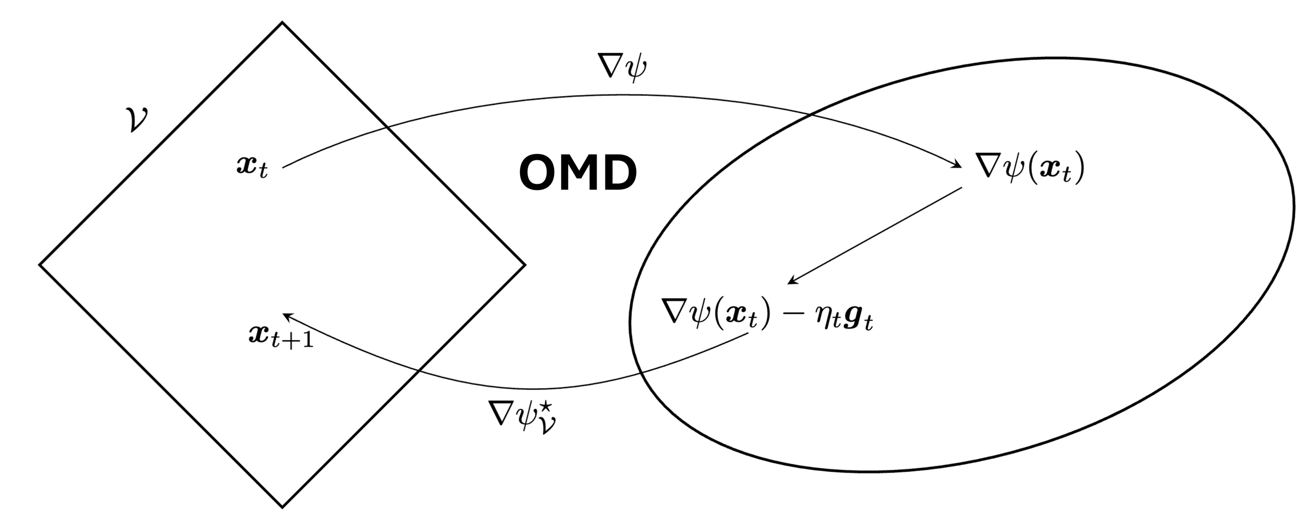
$\boldsymbol{g}_t \in \partial f_t(\boldsymbol{x}_t)$
$\tilde{\boldsymbol{g}}_t \in \partial \tilde{f}_t(\boldsymbol{x}_t)$

$$\mathcal{R}_T = \sum_{t=1}^{T} f_t(\boldsymbol{x}_t) - f_t(\boldsymbol{u}_t)$$

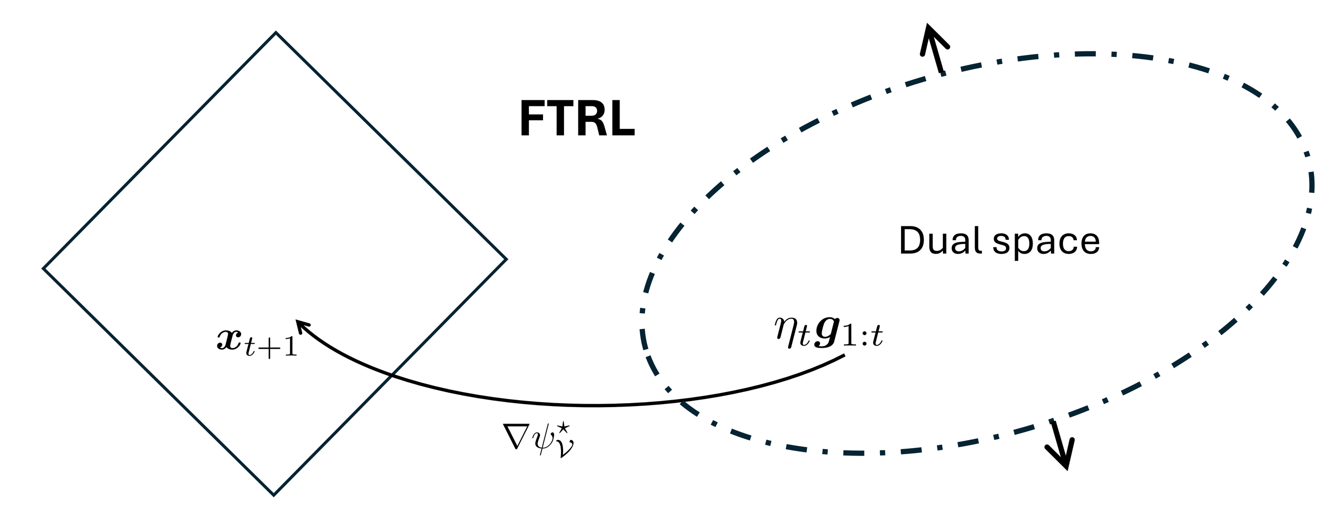
Fact: $\mathcal{R}_T = \Omega\left(\sqrt{TP_T}\right)$ $\quad P_T = \sum_{t=1}^{T-1} \|\boldsymbol{u}_{t+1} - \boldsymbol{u}_t\|$

Aim: $\mathcal{R}_t = \mathcal{O}\left(\sqrt{E_T P_T}\right)$ $\quad E_T = \sum_{t=1}^{T} \epsilon_t^2, \quad \epsilon_t = \|\boldsymbol{g}_t - \tilde{\boldsymbol{g}}_t\|_\star$

2 Main Frameworks



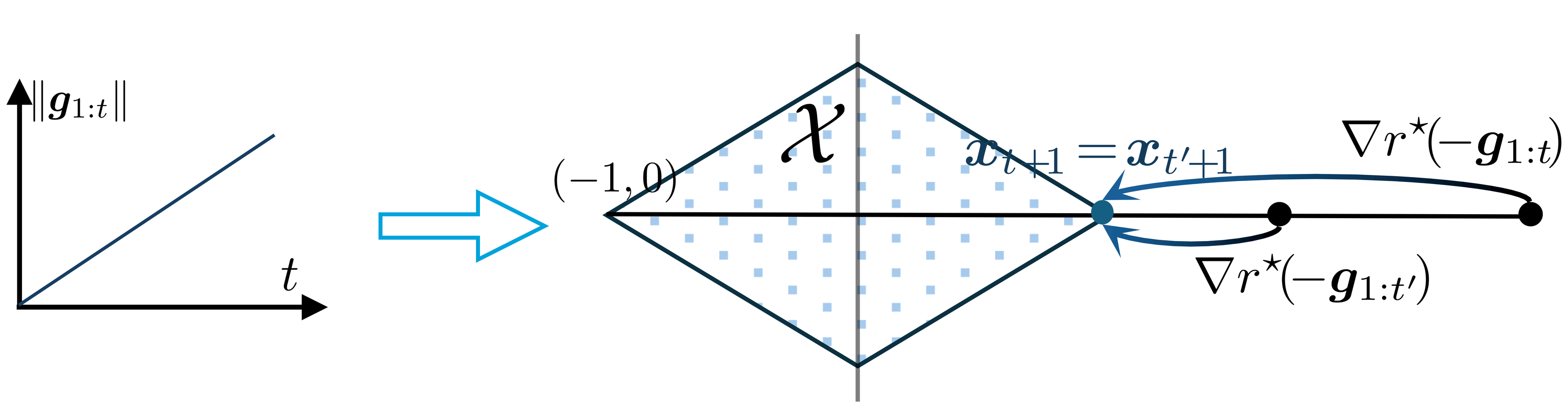
$\mathcal{R}_T = \mathcal{O}\left((P_T + 1)\sqrt{E_T + 1}\right)$ (w/o. ensemble) $\qquad \mathcal{R}_T = \mathcal{O}\left(\sqrt{(P_T + 1)(E_T + P_T)}\right)$ (with ensemble)



Even with constant path ($P_T = 1$) the regret is linear $\mathcal{R}_T = \Omega(T)$.
Why? and how to address this?

3 Insight: Regret Decomposition

$$\mathcal{R}_T \lesssim \mathcal{R}_T^{\text{static}} + \sum_{t=1}^{T-1} \|\boldsymbol{u}_{t+1} - \boldsymbol{u}_t\| \, \|\text{dual state}\|_\star$$



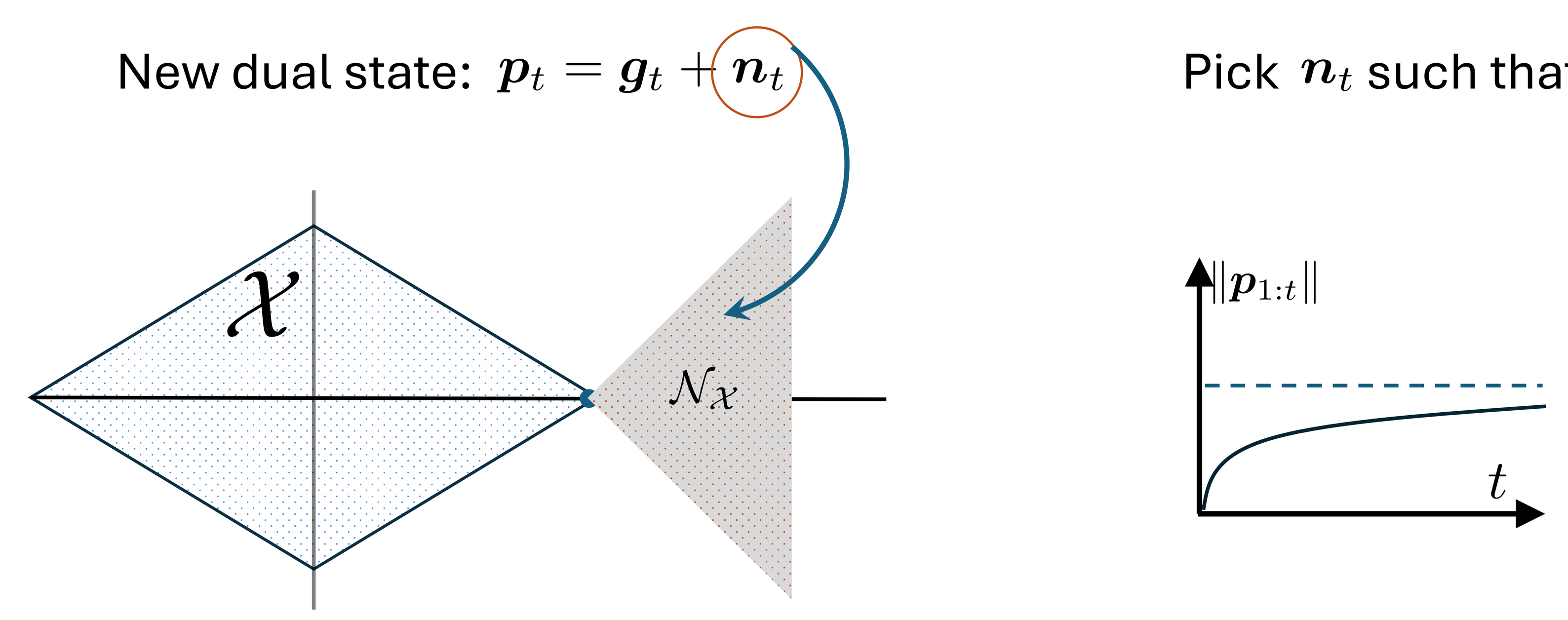
$t < t'$, $g_\tau = (-1, 0)$ for $\tau \le t$, and $g_\tau = (1, 0)$ for $\tau > t$.

Large dual-state obscures the switch.

4 Proposal: Control the Dual State

New dual state: $\boldsymbol{p}_t = \boldsymbol{g}_t + \boldsymbol{n}_t$

Pick $\boldsymbol{n}_t$ such that:

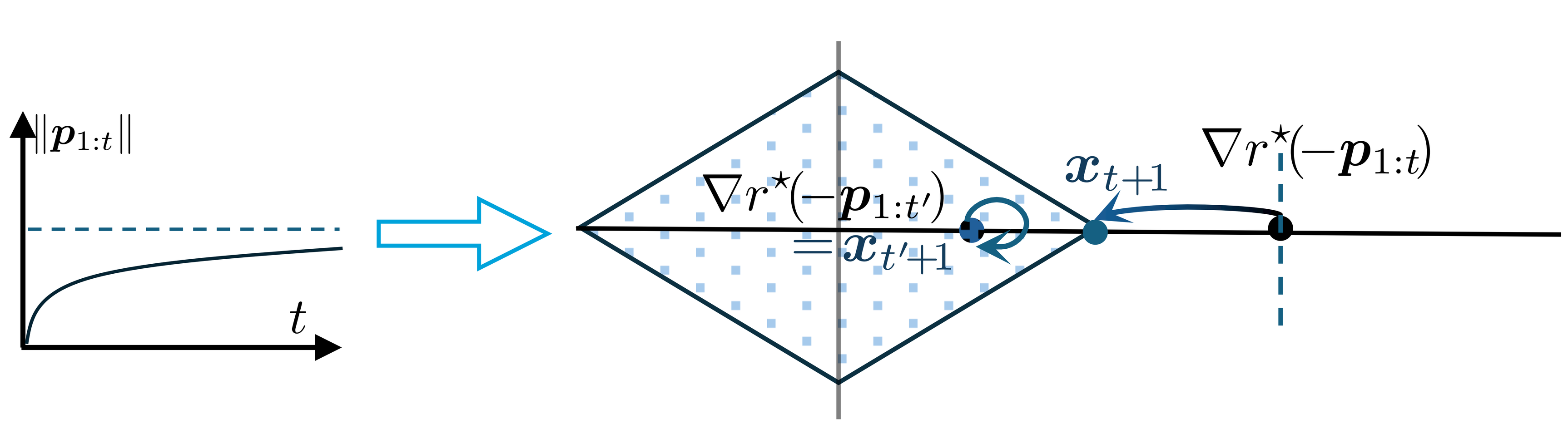


E.g.: $\quad \boldsymbol{x}_t = \arg\min_{\boldsymbol{x} \in \mathcal{X}} \langle \boldsymbol{g}_{1:t-1}, \boldsymbol{x} \rangle + r(\boldsymbol{x}) \Rightarrow \boldsymbol{n}_t = -\boldsymbol{g}_{1:t-1} - \nabla r(\boldsymbol{x}_t)$

Next state: $\quad \boldsymbol{p}_{1:t} = \boldsymbol{g}_t - \nabla r(\boldsymbol{x}_t)$ (recovers OMD's state)

5 OptFPRL: FTRL with Controlled State

$$\boldsymbol{x}_{t+1} = \operatorname*{argmin}_{\boldsymbol{x}} \Big\{ \langle \underbrace{\boldsymbol{p}_{1:t}}_{\text{State vector}}, \boldsymbol{x} \rangle + \underbrace{r_{1:t}(\boldsymbol{x})}_{\text{Incremental regularizers}} + \underbrace{\tilde{f}_{t+1}(\boldsymbol{x})}_{\text{Prediction}} + \underbrace{I_{\mathcal{X}}(\boldsymbol{x})}_{\text{Set constraint}} \Big\}.$$



$$\sum_{t=1}^{T-1} \|\boldsymbol{u}_{t+1} - \boldsymbol{u}_t\| \, \|\text{dual state}\|_\star = \mathcal{O}(\sqrt{E_T}\, P_T + H_T)$$

$$H_T = \sum_{t=1}^{T-1} \|\boldsymbol{u}_{t+1} - \boldsymbol{u}_t\| \, \epsilon_t$$

A bounded dual state induces reactive iterates.

6 OptFPRL Regret

$$\mathcal{R}_T \le (3.7R + P_T)\sqrt{E_T} + H_T = \mathcal{O}\left((1 + P_T)\sqrt{E_T}\right).$$

(w/o. ensemble)

$\mathcal{X} \doteq \{\boldsymbol{x} \in \mathbb{R}^{16} | \, \|\boldsymbol{x}\| \le 2\}$, $f_t(\boldsymbol{x}) = \langle \boldsymbol{c}_t, \boldsymbol{x} \rangle$.

- Sc. 1: $\boldsymbol{c}_{1\to1000} = -1$, $\boldsymbol{c}_{2000\to2500} = -5$, $\boldsymbol{c}_{3500\to3750} = -10$, otherwise $\boldsymbol{c}_t = 1$.
- Sc. 2: $\boldsymbol{c}_t$ alternates between 1 and -1 every 50 steps; $\tilde{\boldsymbol{c}}_t = \boldsymbol{c}_t - \frac{\boldsymbol{c}_t}{0.1t}$.